

Dictionary & Set Comprehensions

Same logic as list comprehensions — now with curly braces {}

■ The Syntax Breakdown

Day 74 is a natural level-up from yesterday. Now that you've mastered list comprehensions [...], switching to Dictionary {key: value} and Set {item} comprehensions is smooth sailing. They use the exact same logic — just with curly braces {}!

1. Dictionary Comprehension

{key: value for item in iterable}

Creates a new dictionary in one line. You define both the key and the value.

```
# Create a dictionary mapping numbers to their squares
numbers = [1, 2, 3, 4]
squares_dict = {x: x**2 for x in numbers}
# Result: {1: 1, 2: 4, 3: 9, 4: 16}
```

2. Set Comprehension

{item for item in iterable}

Creates a set, which automatically removes duplicate values and keeps only unique items.

```
# Extract unique word lengths from a list
words = ["apple", "banana", "apple", "cherry", "banana"]
unique_words = {word.title() for word in words}
# Result: {'Apple', 'Banana', 'Cherry'} (Duplicates automatically gone!)
```

■ Today's Practice Script: day74.py

Create a new file in VS Code: **backend/day74.py** and test these 4 real-world backend scenarios:

Day 74 — Dictionary & Set Comprehensions

```
# =====
# DAY 74: DICTIONARY & SET COMPREHENSIONS
# =====

# 1. Dictionary: Map student names to their passing status (>= 50)
scores = {"Shoaib": 85, "Ali": 42, "Ahmad": 78, "Sara": 35}
pass_status = {name: ("Pass" if score >= 50 else "Fail")
                for name, score in scores.items()}

print("--- 1. Student Pass/Fail Status ---")
print(pass_status)

# 2. Dictionary: Filter user data (Backend API use-case: active users only)
users = [
    {"username": "shoaib", "active": True},
    {"username": "guest1", "active": False},
    {"username": "ahmad", "active": True}
]
active_user_dict = {u["username"]: u["active"] for u in users if u["active"]}

print("\n--- 2. Active Users Dict ---")
print(active_user_dict)

# 3. Set: Extract unique categories from product data (removes duplicates)
products = [
    {"name": "Laptop", "category": "Electronics"},
    {"name": "Shirt", "category": "Apparel"},
    {"name": "Phone", "category": "Electronics"},
    {"name": "Jeans", "category": "Apparel"},
]
unique_categories = {p["category"] for p in products}

print("\n--- 3. Unique Categories Set ---")
print(unique_categories)

# 4. Set: Extract unique tags from a list with spaces/mixed casing
raw_tags = ["python ", "PYTHON", " Django", "django ", "API"]
clean_tags = {tag.strip().upper() for tag in raw_tags}

print("\n--- 4. Clean Unique Tags Set ---")
print(clean_tags)
```

Day 74 — Dictionary & Set Comprehensions

■ Your Task Today

- Create `backend/day74.py` in VS Code and write out the code.
- Run `python day74.py` in your terminal.
- Paste your output here when you're done! ■

■ Dict vs Set — the only difference is the colon

A dictionary comprehension has a `key: value` pair before the `for`. A set comprehension has just a single `item`. Same braces, same loop, same optional `if` filter — just decide whether you're storing pairs or single unique values.